A Formal Refutation of the Square-Root Singularity Conjecture for the Bonacich u-Core

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Abstract

The *Bonacich square-root singularity conjecture* posits that, on sparse Erdős–Rényi random graphs, the order parameter $\phi^*(u)$ of the Bonacich u-core exhibits a mean-field critical exponent, satisfying $\phi^*(u) - \phi_c \sim C\sqrt{u_c - u}$ as $u \nearrow u_c$. We formalize the conjecture in Lean 4 with Mathlib and prove its negation. The refutation is elementary: the identity $\phi - \phi_c = C\sqrt{u_c - u}$, when forced to hold for every $u < u_c$, conflicts with the density bound $\phi^*(u) \leq 1$ inherited from $\phi^*(u) = \lim |S^*(u)|/n$. Specifically, choosing $u = u_c - (2/C)^2$ yields $\phi = \phi_c + 2 > 1$. The result is machine-checked.

1 Introduction

Bonacich centrality quantifies influence in a network by weighting walks of every length with a geometric attenuation parameter β. The associated *u-core*, the maximal vertex set on which every node has centrality at least u, gives a natural hierarchical decomposition of the network. For sparse random graphs $G(n, c/n)$ with $c > 1$ and $\beta < 1/c$, a folkloric mean-field picture predicts a phase transition at some critical threshold u_c, governed by a classical square-root singularity.

We record the conjecture as stated and prove that, *as formalized*, it is false. The obstruction is not subtle: the conjecture requires the *exact* asymptotic identity to hold as an equality for every subcritical u, and this overreach collides with the trivial bound $\phi^*(u) \leq 1$. A corrected statement would require an $\sim$-asymptotic valid only in a neighborhood of u_c, which the formalization does not encode.

Let BSRS denote the conjecture that there exist $u_c > 1$, $\phi_c > 0$, $C > 0$ such that for every $u < u_c$ some $\phi \in (0, 1]$ satisfies $\phi - \phi_c = C\sqrt{u_c - u}$. Then $\neg\,\mathsf{BSRS}$.

2 Formal Statement

The target is the following Lean 4 proposition.

```
def BonacichSquareRootSingularity : Prop :=
      (u_c  _c  C :    ), 1 < u_c      0 <  _c       0 < C
        (u :    ), u < u_c          (   :    ),
      0 <                1         -  _c  = C * Real.sqrt (u_c
          - u)

theorem bonacich_square_root_singularity_neg :
     BonacichSquareRootSingularity
```

3 Natural Language Proof

Suppose, for contradiction, that the conjecture holds. Then there exist constants $u_c > 1$, $\phi_c > 0$, and $C > 0$ such that for every real $u < u_c$ some $\phi \in (0, 1]$ satisfies

$$\phi - \phi_c \;=\; C\sqrt{u_c - u}.$$

Set

$$\delta \;:=\; \left(\frac{2}{C}\right)^2, \qquad u \;:=\; u_c - \delta.$$

Since $C > 0$, we have $\delta > 0$, and therefore $u < u_c$. By hypothesis there exists $\phi \in (0, 1]$ with

$$\phi - \phi_c \;=\; C\sqrt{u_c - u} \;=\; C\sqrt{\delta} \;=\; C \cdot \frac{2}{C} \;=\; 2.$$

Substituting, $\phi = \phi_c + 2$. By assumption $\phi_c > 0$, so $\phi > 2$. This contradicts $\phi \leq 1$. Therefore no such triple (u_c, ϕ_c, C) exists.

4 Formal Lean 4 Proof

The Lean 4 proof mirrors the argument above, using `positivity` for the positivity facts, `Real.sqrt_sq` to evaluate $\sqrt{(2/C)^2} = 2/C$, `field_simp` to simplify $C \cdot (2/C)$, and `linarith` to close the final linear contradiction.

```
theorem bonacich_square_root_singularity_neg :
    BonacichSquareRootSingularity := by
  rintro   u_c  ,  _c , C, hu_c, h _c , hC,  h
  have hC' : (0 :    ) < 2 / C := by positivity
  have h _pos : (0 :    ) < (2 / C) ^ 2 := by positivity
  have hu_lt : u_c - (2 / C) ^ 2 < u_c := by linarith
  obtain      , h _pos , h _le ,  h   _ e q  := h (u_c - (2 /
      C) ^ 2) hu_lt
  have key : u_c - (u_c - (2 / C) ^ 2) = (2 / C) ^ 2 := by
      ring
  rw [key] at h _eq
  rw [Real.sqrt_sq (le_of_lt hC')] at h _eq
  have hne : C     0 := ne_of_gt hC
```

```
have h2 : C * (2 / C) = 2 := by field_simp
rw [h2] at h _eq
linarith
```

5 Conclusion

The Bonacich square-root singularity conjecture, as stated, is false: the density constraint $\phi^*(u) \leq 1$ is incompatible with an unbounded square-root identity required to hold for all subcritical u. A meaningful version of the conjecture must restrict the asymptotic identity to a left neighborhood of u_c. The refutation has been machine-verified in Lean 4 against Mathlib.